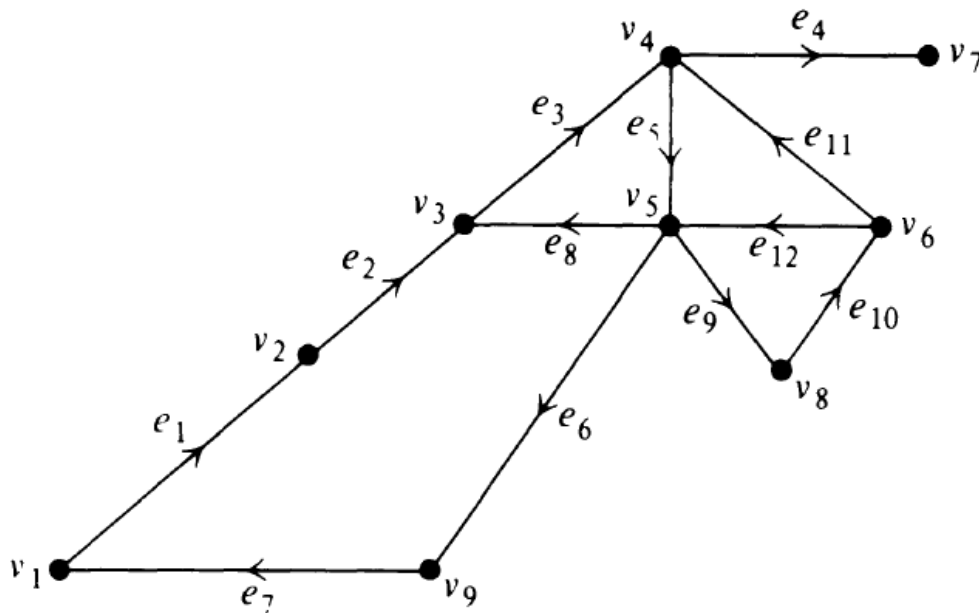


Euler and Hamiltonian Graph

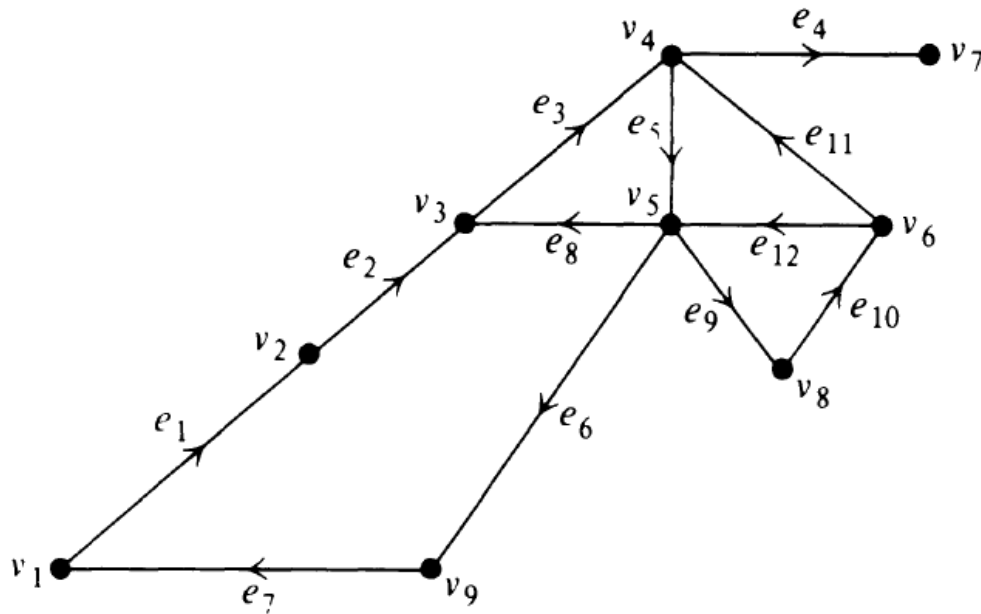
Path

In a directed graph, a *path* is a sequence of edges $(e_{i_1}, e_{i_2}, \dots, e_{i_k})^\dagger$ such that the terminal vertex of e_{i_j} coincides with the initial vertex of $e_{i_{j+1}}$ for $1 \leq j \leq k - 1$. A path is said to be *simple* if it does not include the same edge twice. A path is said to be *elementary* if it does not meet the same vertex twice.



Circuit

A *circuit* is a path $(e_{i_1}, e_{i_2}, \dots, e_{i_k})$ in which the terminal vertex of e_{i_k} coincides with the initial vertex of e_{i_1} . A circuit is said to be *simple* if it does not include the same edge twice. A circuit is said to be *elementary* if it does not meet the same vertex twice.



Theorem 5.1 In a (directed or undirected) graph with n vertices, if there is a path from vertex v_1 to vertex v_2 , then there is a path of no more than $n - 1$ edges from vertex v_1 to vertex v_2 .

PROOF Suppose there is a path from v_1 to v_2 . Let $(v_1, \dots, v_i, \dots, v_2)$ be the sequence of vertices that the path meets when it is traced from v_1 and v_2 . If there are l edges in the path, there are $l + 1$ vertices in the sequence. For l larger than $n - 1$, there must be a vertex v_k that appears more than once in the sequence, that is, $(v_1, \dots, v_i, \dots, v_k, \dots, v_k, \dots, v_2)$. Deleting the edges in the path that leads v_k back to v_k , we have a path from v_1 to v_2 that has fewer edges than the original one. This argument can be repeated until we have a path that has $n - 1$ or fewer edges. □

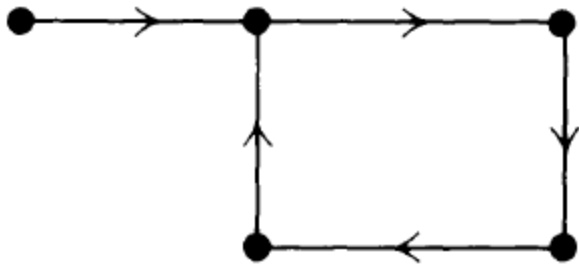
Connected and Disconnected Graph

An undirected graph is said to be *connected* if there is a path between every two vertices, and is said to be *disconnected* otherwise.

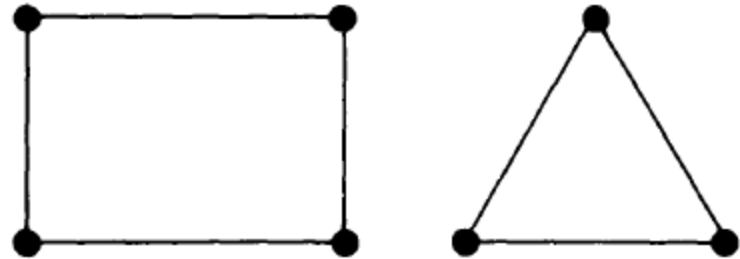
A directed graph is said to be connected if the undirected graph derived from it by ignoring the directions of the edges is connected and is said to be disconnected otherwise.

It follows that a disconnected graph consists of two or more components each of which is a connected graph. A directed graph is said to be *strongly connected* if for every two vertices a and b in the graph there is a path from a to b as well as a path from b to a .

Connected and Disconnected Graph



(a)



(b)

a : Connected Graph

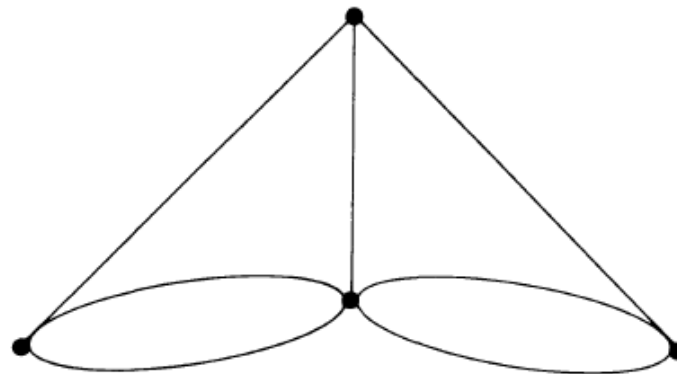
b : Disconnected Graph

Eulerian Paths and Circuits

L. Euler became the father of the theory of graphs when he proved in 1736 that it was not possible to cross each of the seven bridges on the river Pregel in Königsberg, Germany, once and only once in a walking tour. A map of the Königsberg bridges is shown in Fig. 5.14a, and it can be represented as the graph shown in Fig. 5.14b, where the edges represent the bridges and the vertices represent the islands and the two banks of the river.



(a)

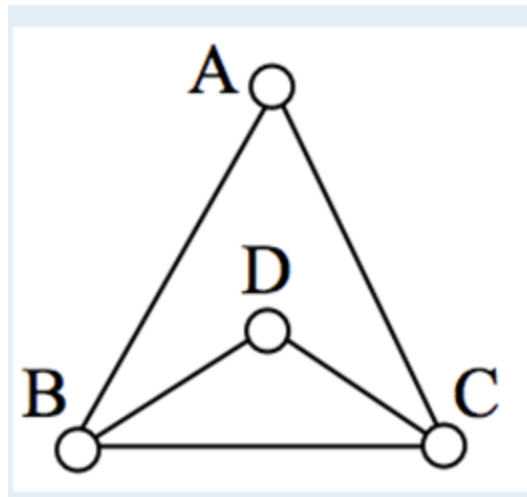


(b)

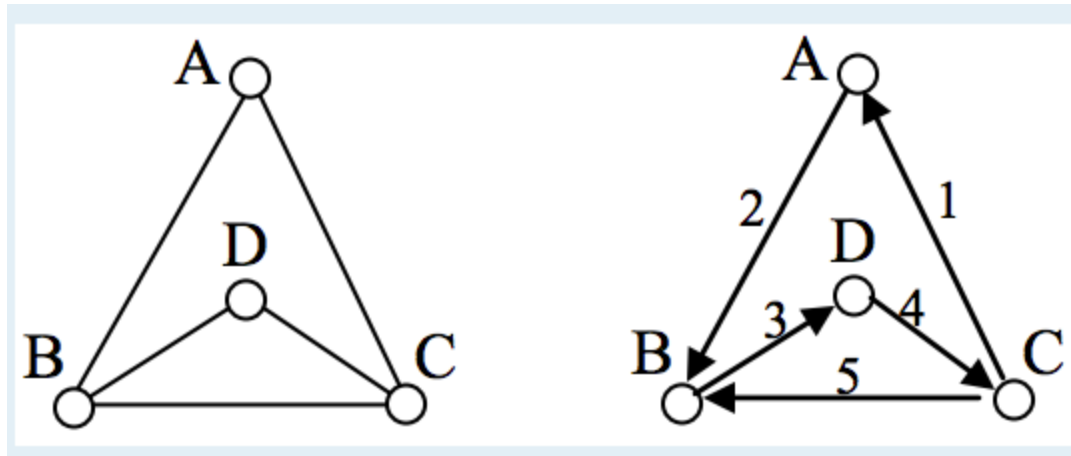
Figure 5.14

Euler Path

- An **Euler path** is a path that uses every edge in a graph with no repeats. Being a path, it does not have to return to the starting vertex.

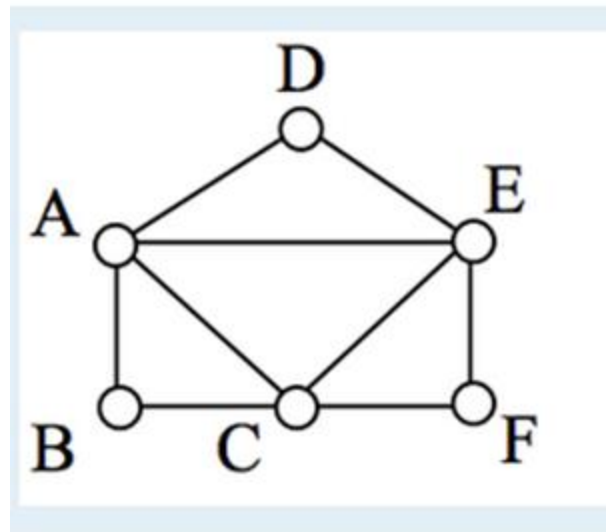


Euler Path

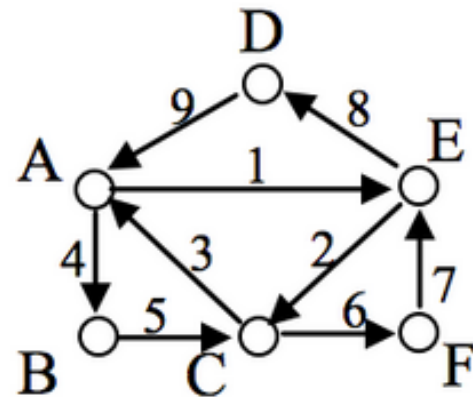
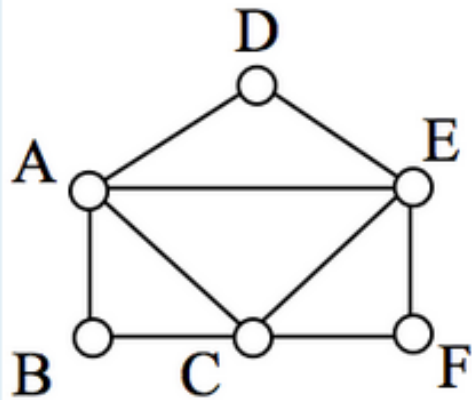


Euler Circuit

- An **Euler circuit** is a circuit that uses every edge in a graph with no repeats. Being a circuit, it must start and end at the same vertex.



Euler Circuit



A result related to Degree of vertices

- In Graph Theory, Handshaking Theorem states in any given graph, Sum of degree of all the vertices is **twice the number of edges contained in it.**

Theorem: Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v)$$

Proof: Each edge contributes twice to the total degree count of all vertices. Thus, both sides of the equation equal to twice the number of edges.

A result related to Degree of vertices

- Given G is an undirected graph, the degree of a vertex v , denoted by $\deg(v)$, in graph G is the number of neighbors of v .
The number of vertices of odd degree in any graph G is even.

Theorem: An undirected graph has an even number of vertices of odd degree.

Proof: Let V_1 be the vertices of even degree and V_2 be the vertices of odd degree in an undirected graph $G = (V, E)$ with m edges. Then

even $\rightarrow$ $2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$

must be
even since
 $\deg(v)$ is
even for
each $v \in V_1$

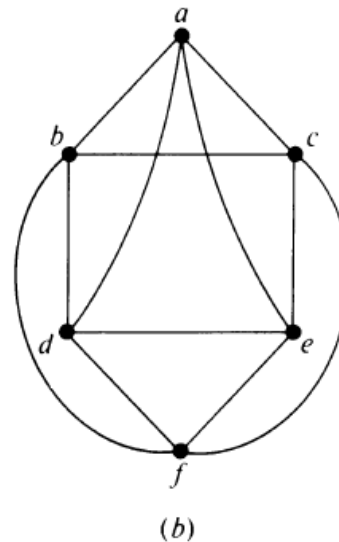
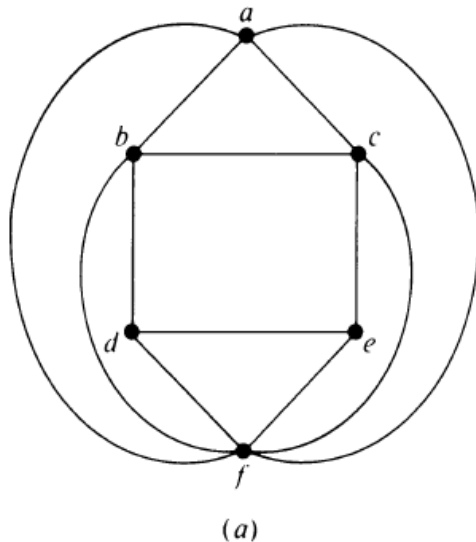
This sum must be even because $2m$ is even and the sum of the degrees of the vertices of even degrees is also even. Because this is the sum of the degrees of all vertices of odd degree in the graph, there must be an even number of such vertices.

Theorems

- **Euler's Path and Circuit Theorems**
 - A graph will contain an Euler path if it contains at most two vertices of odd degree.
 - A graph will contain an Euler circuit if all vertices have even degree

Corollary

- An undirected graph possesses an eulerian circuit if and only if it is connected and its vertices are all of even degree.



Determine above graphs have euler path or circuit based on degree of vertices.

Euler Graph

- Any connected graph is called as an Euler Graph if and only if all its vertices are of even degree.

OR

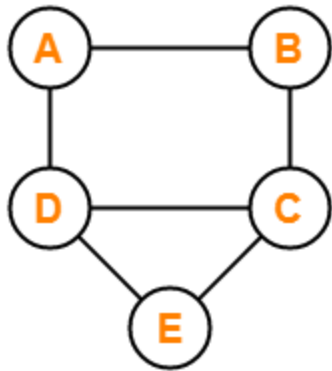
- An Euler Graph is a connected graph that contains an Euler Circuit.

Directed Graph – indegree and outdegree

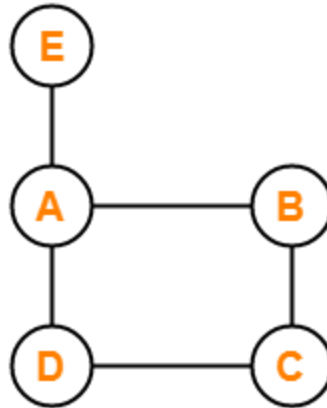
- Indegree – Number of edges incident into the vertex.
- Outdegree – Number of edges incident from the vertex.

Hamiltonian Paths and Circuits

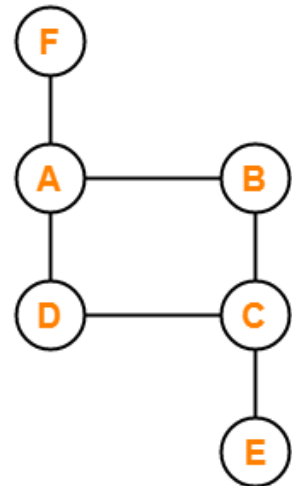
We define a *hamiltonian path* (circuit) to be a path (circuit) that passes through each of the vertices in a graph exactly once.



Hamiltonian Path = ABCDE



Hamiltonian Path = EABCD

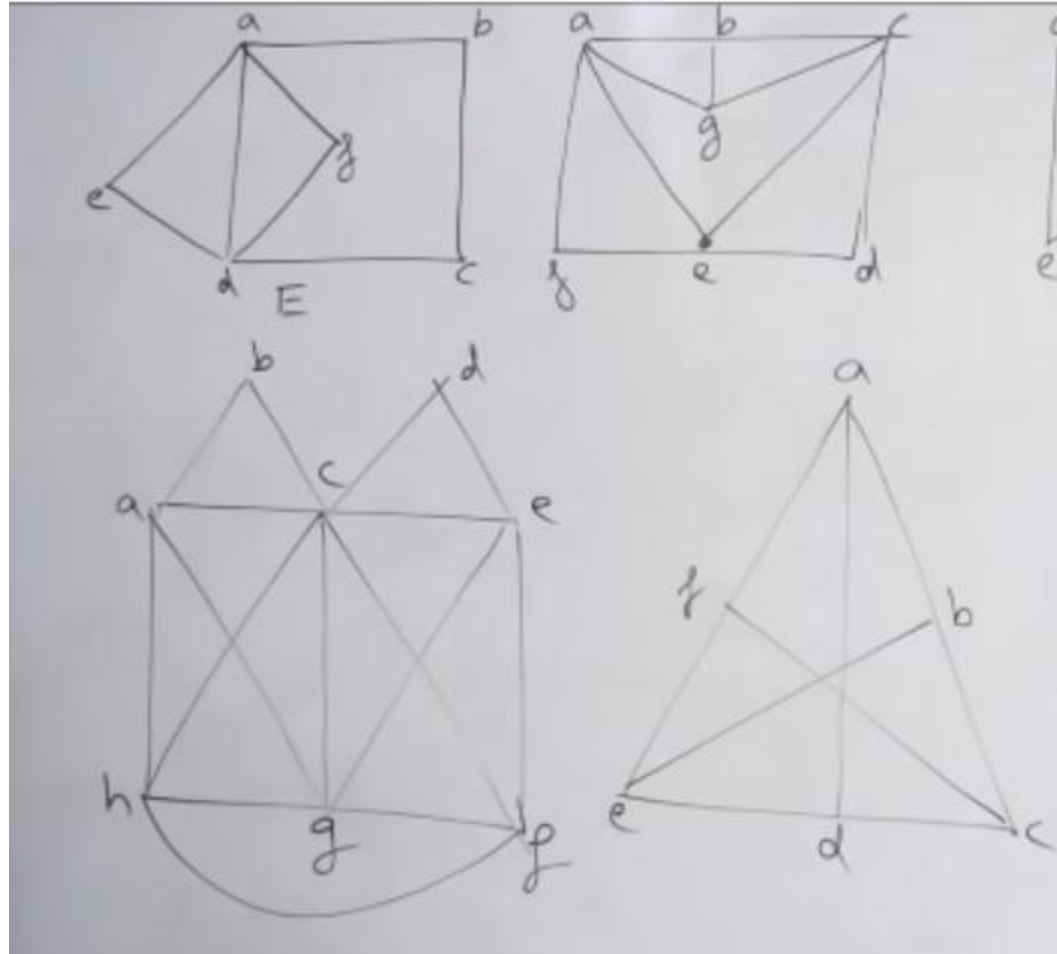


Hamiltonian Path Does Not Exist

Hamiltonian Graph

- Any connected graph that contains a Hamiltonian circuit is called as a Hamiltonian Graph.

Identify : Euler graph and Hamiltonian graph



Identify : Euler graph and Hamiltonian graph

